The Vanishing Violin Caper

Whodunnit? · Functions & Transformations · Teacher Guide and Answer Key

Unit	Time	Supplies	Group size
Functions & Transformations	20–40 minutes	Pencil, scrap paper, scissors	Teams of 2–3 (or solo)

Skills practiced: Function notation, domain and range, transformations of graphs, piecewise functions, composition and inverse functions.

General procedures

1. Give each student the **student sheet** (story and case board) and the **Show Your Work** page.
2. Print the **Clue Cards** (pages 3–7 of the student packet) and cut them apart on the dashed lines. Laminating them lets you reuse the set. Post them around the room.
3. Students visit the cards in any order. At each card they solve the problem on their work page, find their answer among the four choices, read the clue beside it, and cross things off the case board. **Every wrong choice leads to a false clue** that clears the real culprit, so a mistake shows up when the board does not narrow down to one suspect, one place and one method.
4. When a team thinks it knows, they bring you the three names. If they are wrong, let them hunt for the card where something went wrong.

Solution: Rosalind Achebe took it, in the Balcony, using this method: Wrapped in sheet music.

The reveal (read aloud): While the crowd lined up for refreshments, **Rosalind Achebe** slipped up to the **Balcony**, wrapped the Aurelia violin in a stack of **sheet music**, and carried it out under her arm like a review copy. Every alibi and method you cleared pointed straight at her. Bravo, detective!

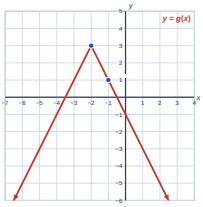
Clue map: what each correct clue clears

Card	Clears
1	Suspect: Celeste Duarte · Place: Orchestra Pit
2	Method: Stuffed in a cello case
3	Suspect: Wen Ling · Place: Tuning Room
4	Method: Swapped for a prop violin
5	Suspect: Fitz Albrecht · Place: Stage Wings
6	Method: Hidden inside the tuba
7	Suspect: Bartholomew Pike · Place: Grand Lobby
8	Method: Tucked under the piano lid
9	Suspect: Jules Navarro · Place: Box Office
10	Method: Rolled in a velvet curtain

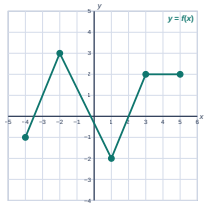
Answer Key

The Vanishing Violin Caper · Correct choice and worked solution for each card

#	Problem	Correct choice	Worked solution	Clue it leads to
1	Evaluate the function at the given input. $f(x) = -2x^2 + 3x - 4$ Find $f(-3)$	**A.** $f(-3) = -31$	Replace every x with (-3), keeping the parentheses: $f(-3) = -2(-3)^2 + 3(-3) - 4$ Square first: $(-3)^2 = 9$, so $-2(9) = -18$. Next, $3(-3) = -9$. $f(-3) = -18 - 9 - 4 = -31$	Celeste Duarte was tuning the cellos in the Orchestra Pit. Forty musicians watched her the whole intermission.
2	Find the domain of the function. Write it in interval notation. $h(x) = \dfrac{\sqrt{(2x - 6)}}{x - 5}$	**B.** $[3, 5) \cup (5, \infty)$	Two rules apply. The expression under a square root must be at least 0, and a denominator can never be 0. $2x - 6 \geq 0$ gives $x \geq 3$. The denominator $x - 5 \neq 0$ gives $x \neq 5$. The value 5 is inside $[3, \infty)$, so remove it. The domain is $[3, 5) \cup (5, \infty)$	Every cello case was opened at the stage door. Inside were only cellos and a few stray rosin cakes.
3	The point $(-3, 8)$ is on the graph of $y = f(x)$. Find the matching point on the graph of g. $g(x) = -2f(x + 1) + 3$	**D.** $(-4, -13)$	x-coordinate: $f(x + 1)$ shifts the graph LEFT 1, so subtract 1 from the x-coordinate: $-3 - 1 = -4$. y-coordinate: multiply by -2 first, then add 3: $-2(8) + 3 = -13$. The matching point is $(-4, -13)$	Wen Ling was warming up in the Tuning Room. A metronome app on her phone logged every minute of it.

4	The graph shows $y = g(x)$, a transformation of the parent function $y = \|x\|$. Write the equation of g. 	**A.** $g(x) = -2\|x + 2\| + 3$	The vertex is $(-2, 3)$, so the graph moved left 2 and up 3: $y = a\|x + 2\| + 3$. From the vertex, move 1 unit right. The graph goes DOWN 2, to $(-1, 1)$. So $a = -2$: a reflection in the x-axis and a vertical stretch by 2. $g(x) = -2\|x + 2\| + 3$	The prop violin was found on its shelf in the original case, dusty and numbered. Nobody swapped it.
5	Use the piecewise function to evaluate the expression. $f(x) = \begin{cases} x^2 - 1 & \text{if } x < 2 \\ 3x - 2 & \text{if } 2 \le x < 5 \\ 20 - 2x & \text{if } x \ge 5 \end{cases}$ $f(-3) + f(2) + f(5)$	**C.** 22	Choose the piece by the input. Since $-3 < 2$, use $x^2 - 1$: $f(-3) = 9 - 1 = 8$. $x = 2$ fits $2 \le x < 5$ (the 2 is included), so use $3x - 2$: $f(2) = 4$. $x = 5$ fits $x \ge 5$, so use $20 - 2x$: $f(5) = 10$. $8 + 4 + 10 = 22$	Fitz Albrecht was resetting the spotlights in the Stage Wings. The lighting console timestamped each cue he ran.
6	Let $f(x) = x^2 - 4x$ and $g(x) = 2x + 3$. Write the composite function as an expression in x. $(f \circ g)(x) = ?$	**D.** $(f \circ g)(x) = 4x^2 + 4x - 3$	$(f \circ g)(x)$ means $f(g(x))$: put $g(x) = 2x + 3$ into f wherever you see x. $f(2x + 3) = (2x + 3)^2 - 4(2x + 3) = 4x^2 + 12x + 9 - 8x - 12$ $(f \circ g)(x) = 4x^2 + 4x - 3$	The tuba was turned upside down and shaken. Out fell a mouthpiece and a peppermint, but no violin.
7	Use the table to evaluate the expression. $(f \circ g)(4) + g^{-1}(3)$	**C.** 7	$(f \circ g)(4) = f(g(4))$. The table gives $g(4) = 5$, and then $f(5) = 2$. $g^{-1}(3)$ asks which input gives $g(x) = 3$. The table shows $g(5) = 3$, so $g^{-1}(3) = 5$. $2 + 5 = 7$	Bartholomew Pike was greeting patrons in the Grand Lobby. The hall photographer has him in every shot.

x	1	2	3	4	5
$f(x)$	3	5	1	4	2
$g(x)$	2	4	1	5	3

8	Find the inverse function. $f(x) = \sqrt[3]{(x - 1)} + 2$	**D.** $f^{-1}(x) = (x - 2)^3 + 1$	Write $y = \sqrt[3]{(x - 1)} + 2$, then swap x and y: $x = \sqrt[3]{(y - 1)} + 2$. Solve for y: $x - 2 = \sqrt[3]{(y - 1)}$. Cube both sides: $(x - 2)^3 = y - 1$. $y = (x - 2)^3 + 1$, so $f^{-1}(x) = (x - 2)^3 + 1$	The piano lid was lifted by the tuner himself. Strings, hammers and felt, but nothing else.
9	A jacket costs \$135. A \$20 coupon is applied first, and then 15% off the reduced price. Let $g(x) = x - 20$ for the coupon and $f(x) = 0.85x$ for the 15% discount. Find the final price. $f(g(135)) = ?$	**B.** \$97.75	The \$20 coupon comes first, so start with g: $g(135) = 135 - 20 = 115$. Taking 15% off means paying 85% of the price: $f(115) = 0.85(115) = 97.75$, so the final price is \$97.75	Jules Navarro was selling programs at the Box Office. The register log shows a sale every few seconds.
10	The graph shows $y = f(x)$. Let $g(x) = 2f(x - 1) + 3$. Find $g(2)$. 	**A.** $g(2) = -1$	Substitute 2 for x: $g(2) = 2f(2 - 1) + 3 = 2f(1) + 3$. On the graph, the point with $x = 1$ is $(1, -2)$, so $f(1) = -2$. $g(2) = 2(-2) + 3 = -1$	The velvet curtain was unrolled and shaken out in front of the crew. It was only dusty fabric.

Wrong-Answer Map

The Vanishing Violin Caper · What a wrong choice looks like, and the false clue it leads to

Card	Wrong choice	Likely error	False clue
1	B. $f(-3) = 5$	treated $(-3)^2$ as -9, so the squared term turned positive	Wen Ling was tuning her violin on the Balcony. The sound crew recorded every note of it.
1	C. $f(-3) = -13$	sign slip on the middle term: $3(-3)$ is -9, not $+9$	Security checked the sheet music folders at the gate. Every page was plain paper, with nothing wrapped inside.
1	D. $f(-3) = 23$	squared -2 together with x, so $(-2x)^2$ instead of $-2(x^2)$	Rosalind Achebe was drafting her review in the Grand Lobby. Three ushers saw her typing at the same table all evening.
2	A. $(3, 5) \cup (5, \infty)$	used > 0 under the root; the radicand can equal 0, so 3 is included (bracket)	Celeste Duarte was greeting donors on the Balcony. A donor's phone video shows her there the whole time.
2	C. $(-\infty, 5) \cup (5, \infty)$	excluded $x = 5$ but ignored the square root: x must also satisfy $2x - 6 \geq 0$	The music librarian counted every score on the cart. No stack was thick enough to hide a violin.
2	D. $[3, \infty)$	checked the square root but forgot that the denominator cannot be 0 at $x = 5$	Rosalind Achebe was interviewing the oboist in the Orchestra Pit. The recording has her questions from start to finish.
3	A. $(-4, 19)$	missed the reflection: the factor is -2, not 2	The program bundles were opened and flipped through. Nothing was rolled up inside any of the pages.
3	B. $(-4, -22)$	did the $+3$ before the -2: the vertical stretch and reflection come first, then the shift	Fitz Albrecht was fixing a spotlight on the Balcony. The work order is signed by the electrician standing beside him.
3	C. $(-2, -13)$	$f(x + 1)$ moves the graph LEFT, so the x-coordinate decreases by 1; this answer moved right	Rosalind Achebe was buying a ticket for the next concert at the Box Office. The receipt is stamped at the exact minute of the theft.
4	B. $g(x) = 2\lvert x + 2\rvert + 3$	missed the reflection: the graph opens down, so the multiplier is negative	Bartholomew Pike was hunting for a dropped tuning fork on the Balcony. Two ushers helped him look the entire intermission.
4	C. $g(x) = -\frac{1}{2}\lvert x + 2\rvert + 3$	misread the stretch: one unit right drops the graph 2 units, so the factor is 2, not one half	A volunteer weighed the box of spare scores. It matched the library list and held no hidden shape.
4	D. $g(x) = -2\lvert x - 2\rvert + 3$	shift sign error: the vertex is at $x = -2$, so the factor is $\lvert x + 2\rvert$	Rosalind Achebe was borrowing a pencil in the Tuning Room. The sign-in sheet has her name in green ink.
5	A. 25	used the middle piece at $x = 5$, but $5 \geq 5$ belongs to the last piece ($x < 5$ excludes 5)	Jules Navarro was counting ticket stubs on the Balcony. The security camera has him there for the entire hour.
5	B. 21	used the first piece at $x = 2$, but $2 \geq 2$ belongs to the middle piece (the 2 is included there)	Rosalind Achebe was chatting with the crew in the Stage Wings. The stage manager logged her visitor badge.
5	D. 4	evaluated $(-3)^2$ as -9 in the first piece	Every stand was checked and the sheet music on it was shaken out. Nothing was tucked between the pages.
6	A. $(f \circ g)(x) = 2x^2 - 8x + 3$	read the composition in the wrong order: this is $(g \circ f)(x) = g(f(x))$	Rosalind Achebe was posing for the hall photographer in the Grand Lobby. Her face is in every picture from intermission.

6	B. $(f \circ g)(x) = 2x^3 - 5x^2 - 12x$	multiplied the two functions instead of putting $g(x)$ inside f	Wen Ling was signing autographs on the Balcony. A line of fans stood with her until the lights came back.
6	C. $(f \circ g)(x) = 4x^2 + 8x + 9$	replaced the first x with $2x + 3$ but forgot to substitute in the $-4x$ term	The orchestra's score cabinet was emptied onto the floor. Only paper and a single paper clip came out.
7	A. 3	read $g(3) = 1$ from the table; $g^{-1}(3)$ needs the input that OUTPUTS 3	Celeste Duarte was reviewing the score on the Balcony. Her assistant's notes have the exact times.
7	B. 10	read the composition backwards: used $g(f(4))$ instead of $f(g(4))$	Rosalind Achebe was asking about subscription seats at the Box Office. The clerk wrote down the whole conversation.
7	D. 9	evaluated $f(4)$ and skipped the inner function g	A page turner looked through the opera folio and found only loose sheets. Nothing was wrapped in the music.
8	A. $f^{-1}(x) = \frac{1}{\sqrt[3]{(x-1)+2}}$	confused $f^{-1}(x)$ with the reciprocal of $f(x)$	The printed encore booklets were bundled and checked. They were flat, light and completely empty.
8	B. $f^{-1}(x) = (x - 2)^3 - 1$	sign error when undoing the -1 inside the root: it becomes $+ 1$ after cubing	Rosalind Achebe was admiring the harp in the Orchestra Pit. A security camera shows her there until the lights came up.
8	C. $f^{-1}(x) = (x + 2)^3 + 1$	sign error when undoing the $+ 2$: subtract 2, not add	Bartholomew Pike was testing the acoustics from the Balcony. His notebook has a reading for every minute.
9	A. $114.75	applied only the 15% discount and skipped the $20 coupon	Fitz Albrecht was hanging a banner over the Balcony rail. Three stagehands held the ladder for him.
9	C. $94.75	applied the discounts in the wrong order: 15% off first, then the $20 coupon, which is $g(f(135))$	Rosalind Achebe was waiting for a backstage tour in the Stage Wings. Two guides stood beside her the entire time.
9	D. $100.00	treated 15% off as $15 off	Police fanned through the whole stack of concert programs. Nothing was folded around a violin.
10	B. $g(2) = 1$	forgot the vertical stretch factor 2 on $f(1)$	Jules Navarro was seating late guests on the Balcony. The usher's clipboard has his initials on every row.
10	C. $g(2) = 7$	$f(x - 1)$ shifts the graph RIGHT, so $g(2)$ uses $f(1)$, not $f(3)$	Rosalind Achebe was testing a viola in the Tuning Room. The luthier's receipt has her signature at that minute.
10	D. $g(2) = 2$	added 3 before multiplying by 2: the 2 multiplies $f(1)$ only, then add 3	The sheet music rolls in the lobby display were unrolled. Each one held only a poster inside.